ACADEMIC CENTRAL

PRACTICE EXAMINATION

UNIVERSITY · 2ND YEAR

CONTROL SYSTEMS

PAPER 1 · PRACTICE A

MARKS: 100 **TIME: 3 hours**

BEng Mechatronics and Robotics Engineering, 2nd year final examination: modelling with transfer functions and block diagrams, time response and steady-state error, stability and the Routh criterion, root locus, frequency response (Bode plots, gain and phase margin), PID control and state-space models.

Written by Academic Central for practice, following control systems modules at South African universities (e.g. University of Pretoria EBB 320 Control Systems: modelling, time and frequency response, stability, root locus, frequency-response design and state space). It is not a university examination paper.

INSTRUCTIONS AND INFORMATION

1. This question paper consists of FIVE questions. Answer ALL the questions.
2. Show ALL your working. Sketches must be neat and labelled; semi-log graph paper may be used for Bode plots.
3. For a second-order system use: %OS = 100 e^(−ζπ/√(1 − ζ²)), Tp = π / (ωn√(1 − ζ²)) and Ts (2%) = 4 / (ζωn).
4. Non-programmable scientific calculators may be used. Give answers correct to TWO decimal places.
5. Number your answers exactly as the questions are numbered.

QUESTION 1: MODELLING

1.1 Explain the difference between an open-loop and a closed-loop control system, and give one example of each from robotics. (4)

1.2 A mass-spring-damper is described by $m\,x'' + c\,x' + k\,x = f(t)$.

1.2.1 Derive the transfer function X(s)/F(s), assuming zero initial conditions. (3)

1.2.2 For m = 2 kg, c = 8 N·s/m and k = 50 N/m, find the natural frequency ωn and the damping ratio ζ. (4)

1.3 Find the transfer function Vout(s)/Vin(s) of an RC low-pass circuit with R = 10 kΩ and C = 100 µF (output across C), and state its time constant. (3)

Answer QUESTIONS 1.4 and 1.5 using FIGURE 1.

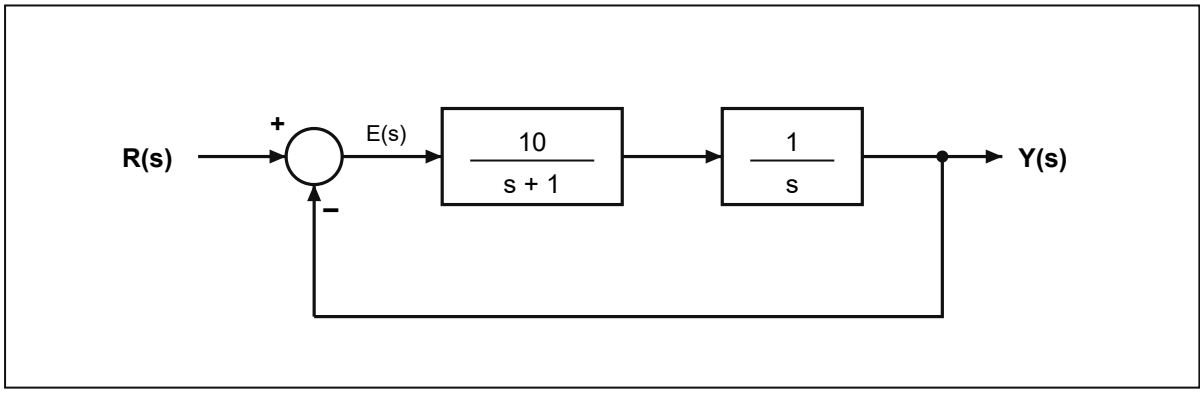


FIGURE 1: Unity-feedback control system

1.4 Reduce the block diagram in FIGURE 1 to a single closed-loop transfer function Y(s)/R(s). (4)

1.5 Write down the poles and the zeros of G(s) = (s + 3) / (s(s + 2)(s + 5)). (2)

[20]

QUESTION 2: TIME RESPONSE

2.1 A first-order system G(s) = 5 / (2s + 1) receives a unit step input. Find the final value, the time constant, the 2% settling time and the output at t = 2 s. (5)

2.2 The closed-loop transfer function of a position servo is $T(s) = 25 / (s^2 + 4s + 25)$. For a unit step input, calculate:

2.2.1 ωn, ζ and the damped natural frequency ωd (3)

2.2.2 the percentage overshoot (2)

2.2.3 the peak time and the 2% settling time (2)

2.2.4 the positions of the closed-loop poles (2)

2.3 For the unity-feedback system in FIGURE 1, find the steady-state error for a unit step input and for a unit ramp input. (4)

2.4 Explain what happens to the overshoot and to the steady-state ramp error of the system in FIGURE 1 if the gain 10 is increased. (2)

[20]

QUESTION 3: STABILITY AND ROOT LOCUS

3.1 Define a stable system, and explain how the positions of the closed-loop poles show whether a system is stable. (3)

3.2 A unity-feedback system has the open-loop transfer function $G(s) = K / (s(s + 1)(s + 5))$.

3.2.1 Use the Routh–Hurwitz criterion to find the range of K for which the closed-loop system is stable. (6)

3.2.2 Find the frequency of oscillation when K is at its upper limit. (3)

3.3 A unity-feedback system has $G(s) = K / (s(s + 4))$.

3.3.1 Sketch the root locus for $K \geq 0$, showing the open-loop poles, the part of the real axis on the locus, the breakaway point and the asymptotes. (4)

3.3.2 Find the value of K at the breakaway point. (2)

3.3.3 Find the value of K that gives a damping ratio of 0.707. (2)

[20]

QUESTION 4: FREQUENCY RESPONSE

An open-loop transfer function is $G(s) = 100 / (s(s + 10))$.

4.1 Sketch the asymptotic Bode magnitude plot of G, stating the corner frequency and the slopes. (4)

4.2 Calculate the magnitude in dB at $\omega = 1$ rad/s and the phase at $\omega = 10$ rad/s. (5)

4.3 Calculate the gain crossover frequency. (3)

4.4 Calculate the phase margin, and state the gain margin. (5)

4.5 Explain what the gain margin and the phase margin tell an engineer about a control system. (3)

[20]

QUESTION 5: CONTROLLERS AND STATE SPACE

5.1 Explain the effect of the proportional, the integral and the derivative terms of a PID controller on the response of a system. (6)

5.2 Why does adding integral action remove the steady-state error to a step input for a type 0 plant? (2)

5.3 Write the system $x'' + 4x' + 25x = u$, with output $y = x$, in state-space form (find A, B, C and D). (5)

5.4 Find the eigenvalues of A in 5.3 and compare them with the poles in QUESTION 2.2. (3)

5.5 A PID controller runs on a microcontroller. Why must it run at a fixed sampling time, much faster than the system responds? (4)

[20]

TOTAL: 100

END OF QUESTION PAPER